MemSkill: Learning and Evolving Memory Skills for Self-Evolving Agents

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HarnessRSI 70 ★★★

evidence: improver

Paper

Code

1 What changes, what is kept

The kept skill bank is chosen by training-split reward (Eq. 8), not a held-out score. What the LoCoMo validation split is used for is not stated.

Skill bank: skills are text guides that get edited

Capture Temporal Context

Purpose: what it captures
When to use: trigger condition
How to apply: steps
Constraints: what to avoid

Starts with 4 primitives

INSERT · UPDATE · DELETE · SKIP

App. C lists 9 / 7 skills (representative)

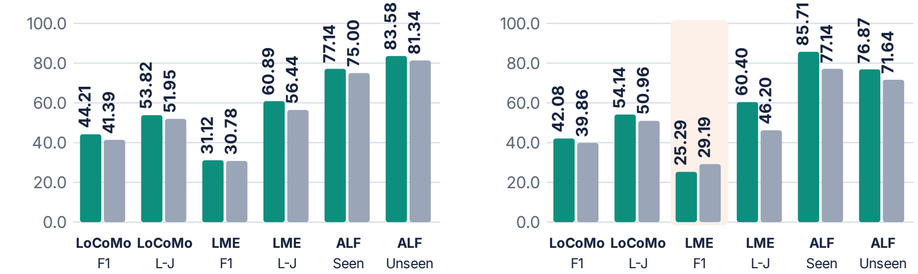
1. **Controller picks Top-K skills** — 3 small MLPs (PPO) score every skill against the span; K = 3 in training, 7 or 5 at test
2. **Executor LLM writes memory** — one call per text span applies the selected skills and updates the memory bank
3. **Reward from training questions** — memory answers train queries: F1 / success; failures fill a hard-case buffer
4. **Designer LLM edits the bank** — every 100 steps: at most 3 edits, refine a skill or add one (model not named)
5. **Keep, or roll back** — best skill-bank snapshot by tail training reward; no gain: roll back

loop

best skill-bank snapshot is kept
base LLM weights never change; controller on rollback: not stated

2 Headline: ahead in 11 of 12 plotted columns

MemSkill · Strongest baseline in that column



One run per cell, no SD; one judge (gpt-oss-120b). Baseline = highest the paper reports: LoCoMo L-J LLaMA LightMem (Table 3), Qwen ALF Unseen Expel (Table 4). LME = LongMemEval; ALF with demos. Shaded: not best.

3 Controls that decide if the gain is real

LoCoMo L-J (single runs)	LLaMA	Qwen
MemSkill (full)	53.82	54.14
w/o controller (random skills)	48.43	42.84
w/o designer (4 primitives)	46.50	36.15
refine-only (no new skills)	47.45	48.88
best baseline (LightMem / CoN)	51.95	50.96

All ablations drop (on LLaMA also below A-MEM 49.71). Split not stated.

Transfer · LLaMA, skills from LoCoMo (Fig. 3)

HotpotQA L-J	K = 7	K = 3	MemoryOS	A-MEM
50 docs	67.57	64.06	66.02	65.63
100 docs	71.48	68.75	67.19	66.80
200 docs	67.97	59.76	60.55	61.72

Only MemoryOS, A-MEM shown; CoN (conv. avg 49.08, above both) and LightMem omitted. Only K = 7 beats both at every length.

Cost · LLaMA, LoCoMo, runtime only (Table 3)

LoCoMo runtime	L-J	In (K)	Calls
MemSkill (span 512)	53.82	249	215
LightMem	51.95	789	685
A-MEM	49.71	2850	1548

Not reported: seeds / variance, matched compute, baseline tuning, evolution cost, leakage test. Span 512 = best of 5 sizes on the same run.

4 Where the gain is absent or thin

- **Qwen LongMemEval F1: 25.29**, below CoN 29.19 and Mem0 27.36 (L-J 60.40 is ahead)
- Thin: LoCoMo L-J 53.82 vs LightMem 51.95; ALF Unseen 83.58 vs 81.34; AppWorld with demos 47.32 vs AWM 47.11, No-Memory 46.41
- HotpotQA at K = 3: below both baselines (50 / 200 docs)
- A.1 "strongest baseline" = added ones; vs Table 1 Mem0 / CoN ALF avg margins: **2.54‡** / **7.27‡**
- Small test sets: LoCoMo about 2 conversations (inference), LongMemEval about 100 questions; no CIs

Takeaway

Learning which memory skills to apply, and letting an LLM rewrite them, beats eight baselines in most columns, and each ablation lowers it. But every number is one run on a small test set, what the baselines are tuned with is not stated, evolution cost is not reported, and some columns show no gain.

Our read · HarnessRSI

Score 70 · ★★★ · evidence level: improver

Reports: transfer, ablations, runtime cost.
Does not report: variance, matched compute, evolution cost, leakage test.